Siyuan (Wilson) Sun

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Education

University of Arizona — Tucson, AZ
B.S. in Artificial Intelligence; GPA: 3.85/4.0 — *Aug. 2025 – May 2027 (Expected)*
- **Relevant Coursework:** Deep Learning for NLP, Neural Networks, Reinforcement Learning, Principles of ML, Text Retrieval and Web Search.

East China University of Science and Technology — Shanghai, China
B.S. in Artificial Intelligence (Transferred to UArizona) — *Sep. 2023 – July 2025*

Research Experience

LUMIA Lab, Shanghai Jiao Tong University — Shanghai, China
Research Intern (Advised by Prof. Zhouhan Lin) — *Apr. 2026 – Present*
- Researching **parametric memory for LLMs**, LLM pretraining, and test-time training (TTT).

CLULab, University of Arizona — Tucson, AZ
Undergraduate Research Assistant (Advised by Prof. Mihai Surdeanu) — *Sep. 2025 – Present*
- Researching **LLMs and Information Retrieval**, with a focus on Transformer-based neural retrieval systems and query expansion integration.

Technical Skills

Programming: Python, C/C++, Java, Bash.
Deep Learning & LLMs: PyTorch, Hugging Face Transformers, Accelerate, PEFT (LoRA).
Memory & Retrieval Methods: Parametric memory, RAG, dense retrieval, contrastive learning, hard-negative mining, query expansion, hybrid retrieval, reranking.
Retrieval Tools: FAISS, Pyserini, Sentence Transformers, Elasticsearch, Milvus.
Training & Optimization: Large-scale multi-GPU and multi-node distributed training; DDP, FSDP, DeepSpeed ZeRO; mixed-precision training (FP16/BF16).

Publications

Deep Fusion Memory: A Chunk-Level Parametric Memory for Large Language Models

Rubin Wei, **Siyuan Sun**, Hao Sun, Yuliang Liu, Xiang Shi, Jiarui Wang, Junming Zhang, and Zhouhan Lin

2026. ICLR 2027 (Under Review) | https://arxiv.org/abs/2610.25100
- Learns chunk-level memory and fuses it into a frozen LLM via gated cross-attention; no explicit retrieval at inference.
- Improved average scores over the strongest baseline by **12.71% / 11.19%** on six QA tasks (Mistral-7B / Qwen3-8B); domain adaptation gains of **6.51 / 4.38 points** on Qwen3-4B / 8B.

Query Expansion Is More Than Generation: Improving Dense Retrieval through Better Integration

Siyuan Sun and Mihai Surdeanu

2026. WSDM 2027 (Under Review) | https://arxiv.org/abs/2608.25521
- Separately encodes queries and expansions, then interpolates their vectors using an unsupervised online weight estimate.
- Improved retrieval effectiveness by up to **12.89%** across TREC-DL, LoTTE, and BEIR without task-specific training; online calibration beat development-tuned fixed weights by up to **3.81%**.

Selected Projects

Learning Dense Reward Functions in Chess | *Gymnasium, ResNet, IRL*
- Inferred dense rewards from expert chess demonstrations with a custom Gymnasium environment and ResNet reward model; saliency analysis revealed human-like positional patterns.

Brain-to-Text: Decoding Intracortical Speech | *BERT, GPT-2, LoRA* — Brain-to-Text '25
- Built a BERT–GPT-2 neural speech decoder using masked phoneme modeling and LoRA, achieving **12.4% WER**; a day-specific adaptation layer handled non-stationary neural signals.